ASSIGNMENT NO. 4: Working with Multiple Tables and Joins

Questions:

Q1) Create a MySQL database named CollegeDB and create the following tables:

Student Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	
City	VARCHAR(30)	

Course Table

Column Name	Data Type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	
Fees	DECIMAL(10,2)	

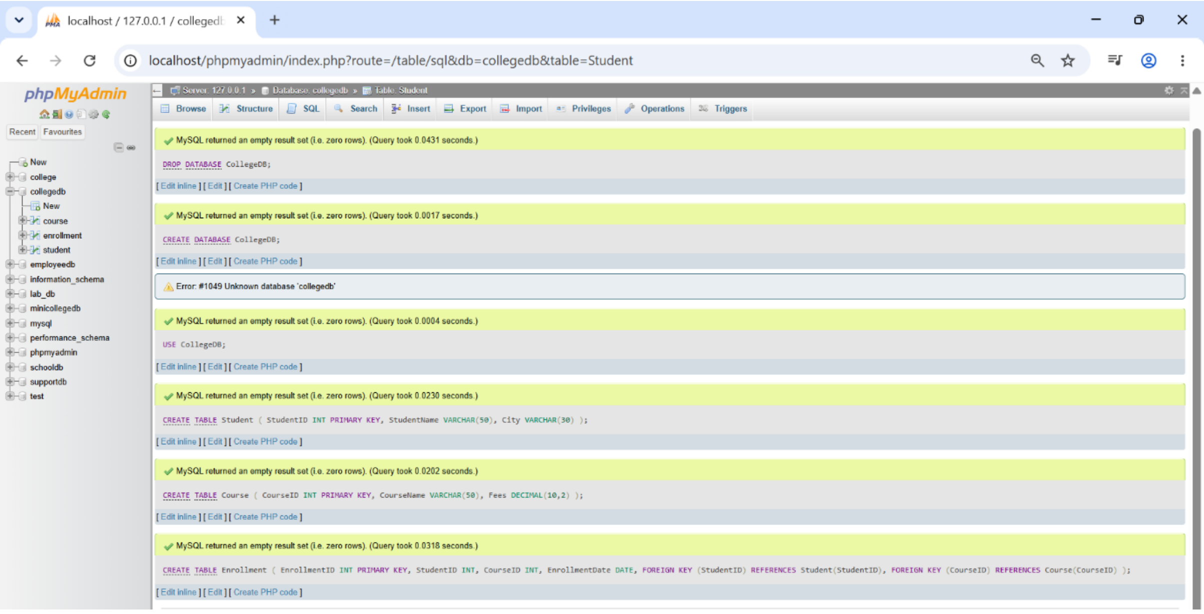
Enrollment Table

Column Name	Data Type	Constraint
EnrollmentID	INT	PRIMARY KEY
StudentID	INT	FOREIGN KEY
CourseID	INT	FOREIGN KEY
EnrollmentDate	DATE	

Query-

CREATE DATABASE CollegeDB; USE CollegeDB;
CREATE TABLE Student (StudentID INT PRIMARY KEY,
CREATE TABLE Student (StudentID INT PRIMARY KEY,
StudentName VARCHAR(50), City VARCHAR(30));
CREATE TABLE Course (CourseID INT PRIMARY KEY,
CourseName VARCHAR(50), Fees DECIMAL(10,2));
CREATE TABLE Enrollment (
EnrollmentID INT PRIMARY KEY,
StudentID INT,
CourseID INT,
EnrollmentDate DATE,
FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
FOREIGN KEY (CourseID) REFERENCES Course(
CourseID));

Output-



Q2) Insert at least 6 records into the Student table, 4 records into the Course table, and 5 records into the Enrollment table.

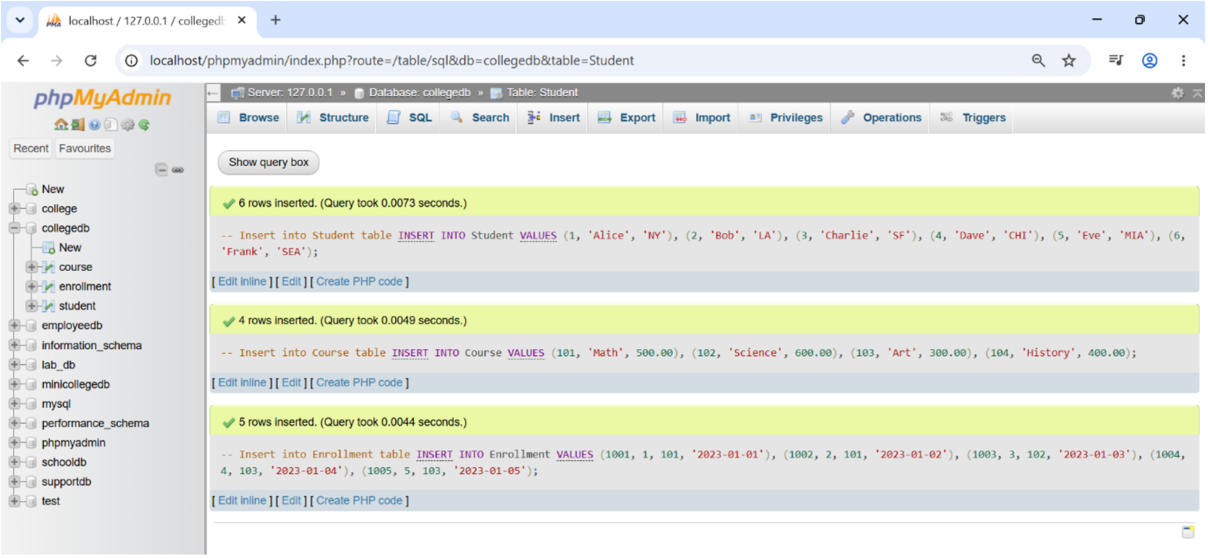
Make sure:

Requirement
At least one student should not have enrollment
At least one course should not have any enrolled student

Query-

INSERT INTO Student VALUES (1, 'Alice', 'NY'),
(2, 'Bob', 'LA'), (3,'Charlie','SF'),
(4, 'Dave', 'CHI'), (5, 'Eve', 'MIA'),
(6, 'Frank', 'SEA');
INSERT INTO Course VALUES
(101, 'Math',26,500.00), (101, 'Math',500.00),
(102,'Science',600.00), (103, 'Art',300.00),
(104, 'History',400.00);
INSERT INTO Enrollment VALUES
(1001,1,101, '2023-01-01'),
(1002,2, 101, '2023-01-02'),
(1003,3,102, '2023-01-03'),
(1004,4,103, '2023-01-04'),
(1005,5,103, '2023-01-05');

Output-



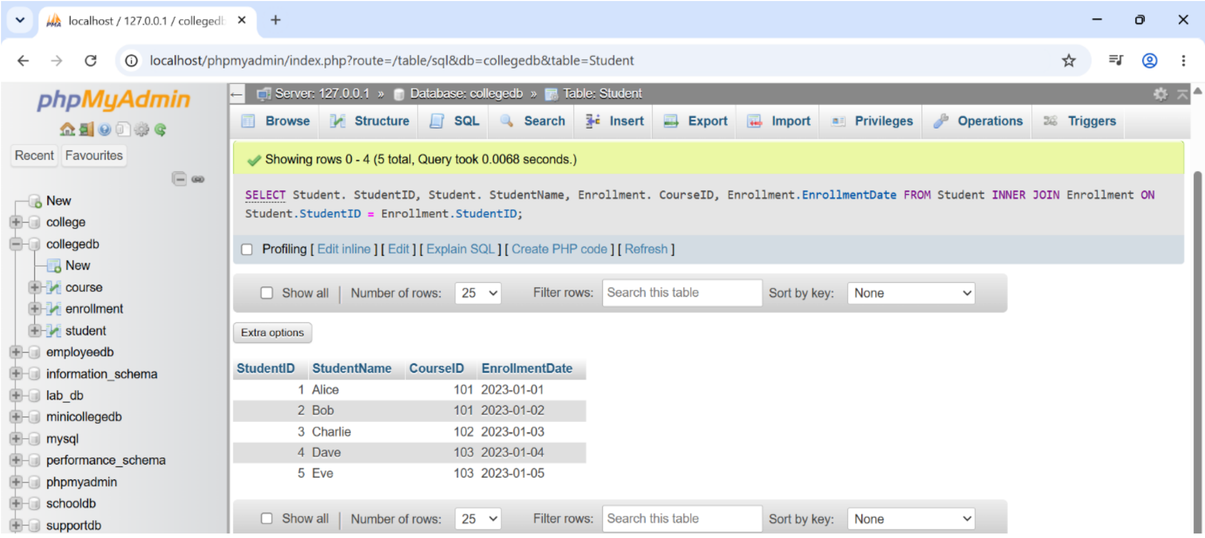
Q3) Write a MySQL query using INNER JOIN to display students who have enrolled in courses.
The output should include:

Column
StudentID
StudentName
CourseID
EnrollmentDate

Query-

SELECT Student. StudentID,
Student. StudentName,
Enrollment. CourseID,
Enrollment.EnrollmentDate
FROM Student
INNER JOIN Enrollment ON
Student.StudentID =
Enrollment.StudentID;

Output-



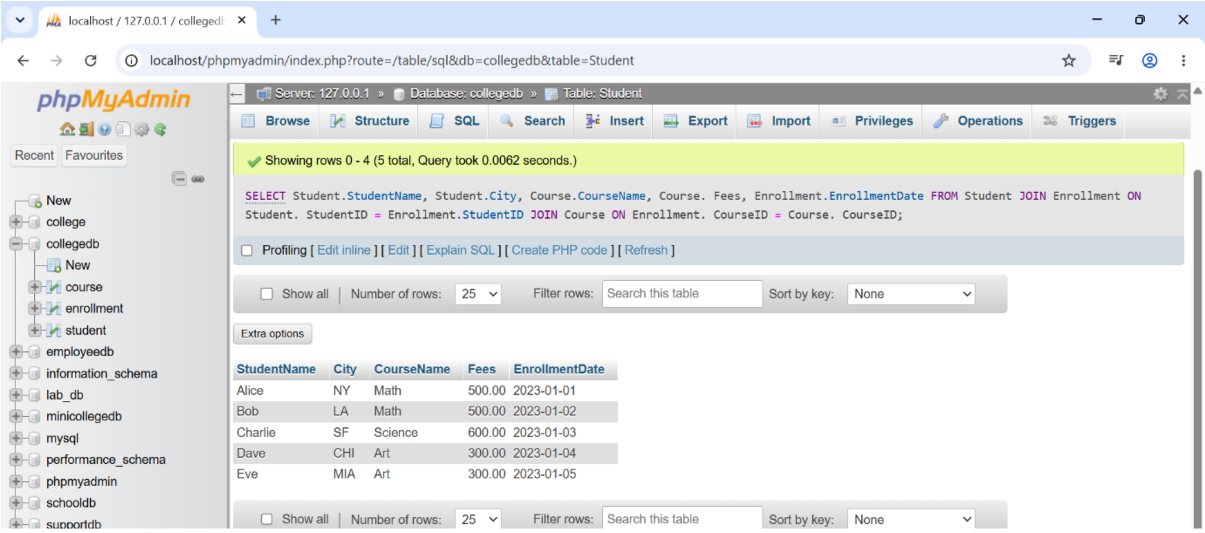
Q4) Write a MySQL query to display student name, city, course name, fees, and enrollment date using multiple tables.
Use:

Tables Required
Student
Course
Enrollment

Query-

SELECT Student.StudentName,
Student.City,
Course.CourseName,
Course. Fees,
Enrollment.EnrollmentDate
FROM Student
JOIN Enrollment ON
Student. StudentID =
Enrollment.StudentID
JOIN Course ON Enrollment. CourseID
= Course. CourseID;

Output-



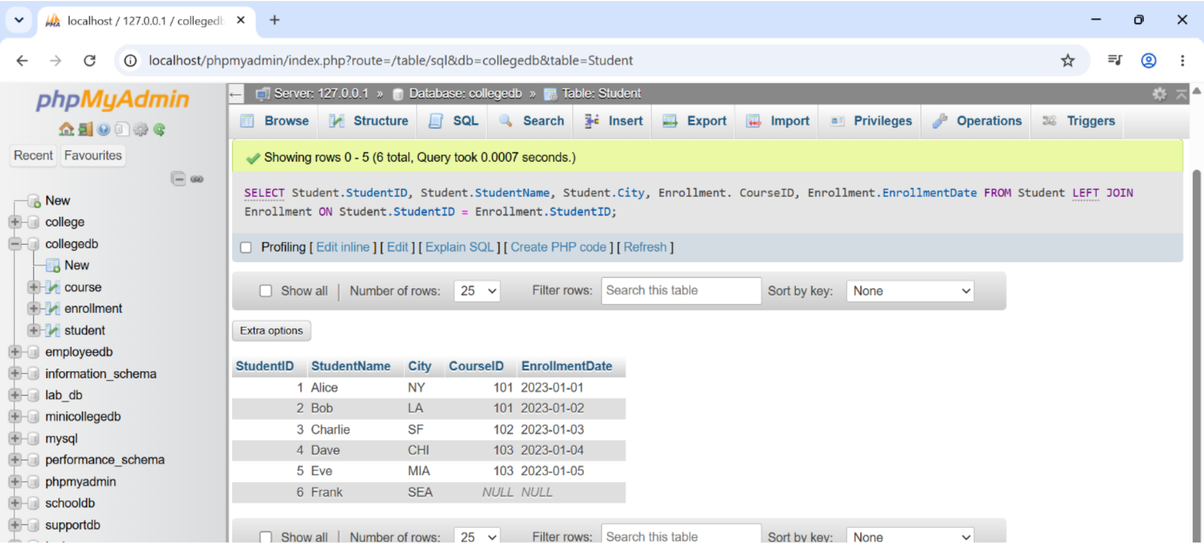
Q5) Write a MySQL query using LEFT JOIN to display all students, including those who have not enrolled in any course.
The output should include:

Column
StudentID
StudentName
City
CourseID
EnrollmentDate

Query-

SELECT Student.StudentID,
Student.StudentName,
Student.City,
Enrollment. CourseID,
Enrollment.EnrollmentDate
FROM
Student
LEFT JOIN Enrollment
ON Student.StudentID = Enrollment.StudentID;

Output-



Q6) Write a MySQL query using RIGHT JOIN to display all courses, including courses that have no enrolled students.

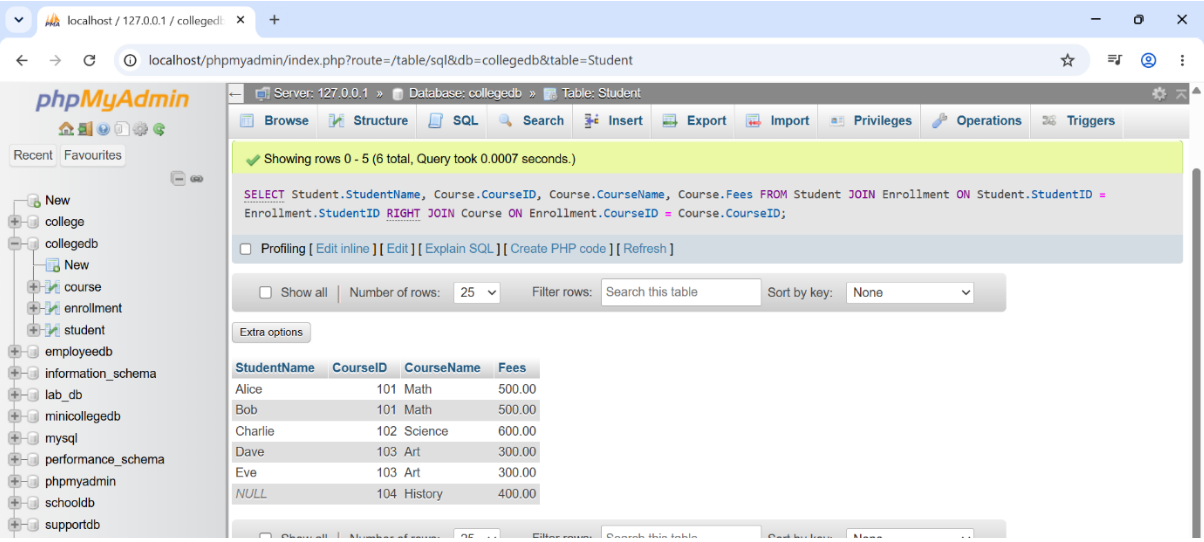
The output should include:

Column
StudentName
CourseID
CourseName
Fees

Query-

SELECT Student.StudentName,
Course.CourseID,
Course.CourseName,
Course.Fees
FROM Student
JOIN Enrollment ON
Student.StudentID = Enrollment.StudentID
RIGHT JOIN Course ON
Enrollment.CourseID = Course.CourseID;

Output-


localhost / 127.0.0.1 / colleged
localhost/phpmyadmin/index.php?route=/table/sql&db=collegedb&table=Student
phpMyAdmin
Recent Favourites
New
college
collegedb
New
course
enrollment
student
employeedb
information_schema
lab_db
minicollegedb
mysql
performance_schema
phpmyadmin
schooldb
supportdb
Server: 127.0.0.1 » Database: collegedb » Table: Student
Browse Structure SQL Search Insert Export Import Privileges Operations Triggers
Showing rows 0 - 5 (6 total, Query took 0.0007 seconds.)
SELECT Student.StudentName, Course.CourseID, Course.CourseName, Course.Fees FROM Student JOIN Enrollment ON Student.StudentID = Enrollment.StudentID RIGHT JOIN Course ON Enrollment.CourseID = Course.CourseID;
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
StudentName CourseID CourseName Fees
Alice 101 Math 500.00
Bob 101 Math 500.00
Charlie 102 Science 600.00
Dave 103 Art 300.00
Eve 103 Art 300.00
NULL 104 History 400.00